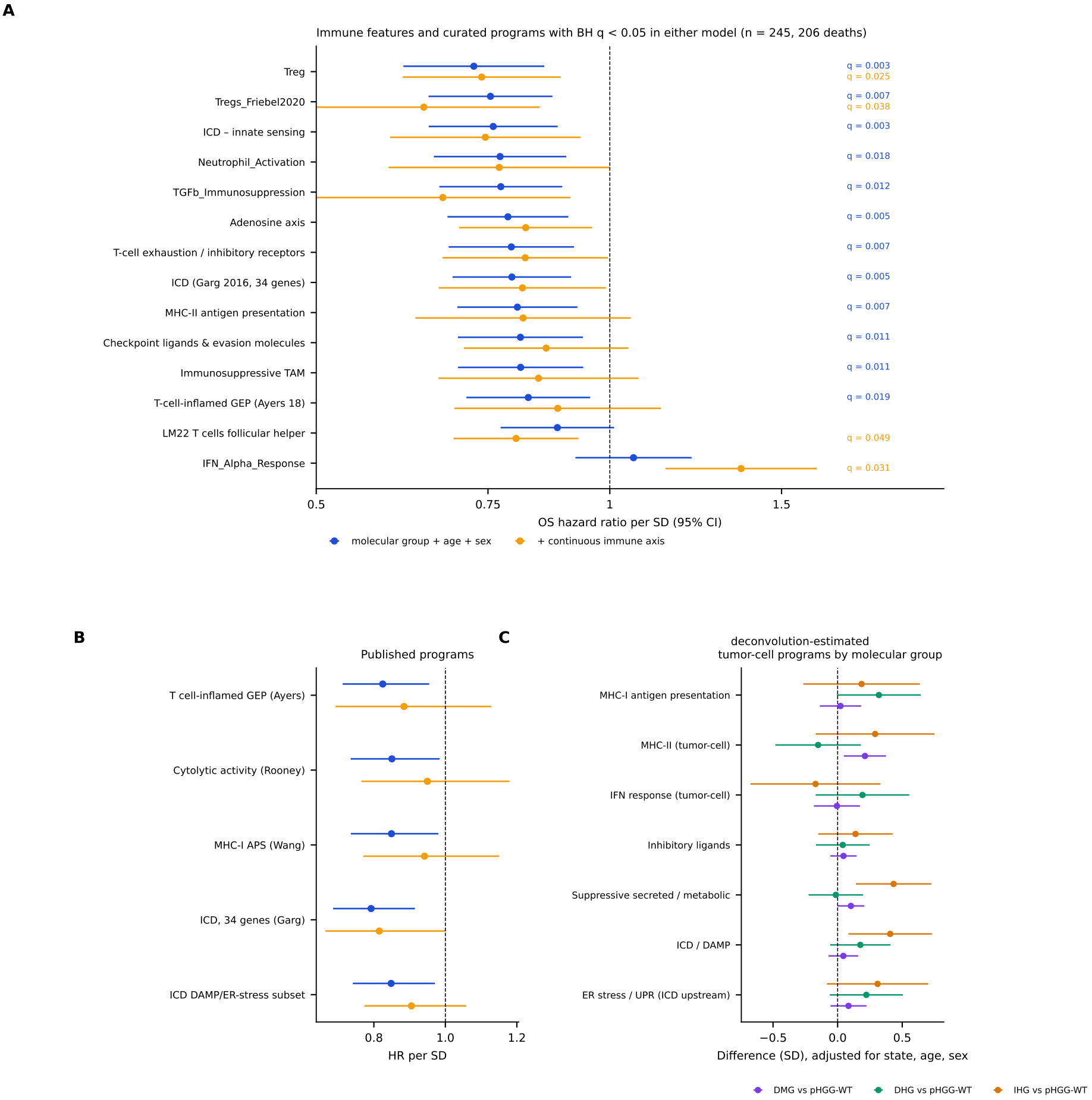

A
Immune features and curated programs with BH q < 0.05 in either model (n = 245, 206 deaths)
Treg
Tregs_Friebel2020
ICD – innate sensing
Neutrophil_Activation
TGFb_Immunosuppression
Adenosine axis
T-cell exhaustion / inhibitory receptors
ICD (Garg 2016, 34 genes)
MHC-II antigen presentation
Checkpoint ligands & evasion molecules
Immunosuppressive TAM
T-cell-inflamed GEP (Ayers 18)
LM22 T cells follicular helper
IFN_Alpha_Response
q = 0.003
q = 0.025
q = 0.007
q = 0.038
q = 0.003
q = 0.018
q = 0.012
q = 0.005
q = 0.007
q = 0.005
q = 0.007
q = 0.011
q = 0.011
q = 0.019
q = 0.049
q = 0.031
0.5
0.75
1
1.5
OS hazard ratio per SD (95% CI)
molecular group + age + sex
+ continuous immune axis
B
Published programs
T cell-inflamed GEP (Ayers)
Cytolytic activity (Rooney)
MHC-I APS (Wang)
ICD, 34 genes (Garg)
ICD DAMP/ER-stress subset
0.8
1.0
1.2
HR per SD
C
deconvolution-estimated
tumor-cell programs by molecular group
MHC-I antigen presentation
MHC-II (tumor-cell)
IFN response (tumor-cell)
Inhibitory ligands
Suppressive secreted / metabolic
ICD / DAMP
ER stress / UPR (ICD upstream)
−0.5
0.0
0.5
Difference (SD), adjusted for state, age, sex
DMG vs pHGG-WT
DHG vs pHGG-WT
IHG vs pHGG-WT